

Study of Water Resources and Delineation of Watershed in and Around Pulivendula and Sanivaripalli Area Using Remote Sensing and GIS

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Abstract: Rapid industrial development, urbanization and increase in agricultural production have led to freshwater shortages in many parts of the world. In view of increasing demands of water for various purposes like agricultural, domestic and industrial, a greater emphasis is being laid for a planned and optimal utilization of water resources. On other hand, continuous withdrawals from a groundwater reservoir in excess of replenishable recharge may result in lowering of water table. There is an urgent need of planned and optimal development of water resources. An appropriate strategy is required to develop water resources with planning based on conjunctive use of surface and subsurface water resources. Remote Sensing, with its advantage of spatial, spectral and temporal availability of data covering large and inaccessible area within short time, has become a very rapid and cost-effective tool in assessing, monitoring and conserving groundwater resources. Geographical Information System (GIS) is a powerful set of tools for collection, storing, retrieving, transforming and displaying spatial data from the real world for a particular purpose. Watershed is a geo hydrological unit or piece of land that drain at a common point. The watershed management implies the judicious use of all the resources i.e. land, water, vegetation in an area. The task of watershed management includes the treatment of land by using most suitable geological and engineering measures in such a manner that, the management work must be economic and socially acceptable. For hydrologic analysis of the watershed, the physical structure (catchment factors such as topography, shape, size, soil type, slope and land use) and geometry (morphometric parameters such as linear aspects and drainage aspects) is of prime importance. The ground water distribution in the study area is not uniform and it depends on the drainage pattern. The unique river in the study area is Maddaleru River, a seasonal river fed by monsoon, which is flowing in the NW direction in the southern part. A watershed, near the village Krishnamgaripalli is demarcated so as to fulfill the needs. The total drainage area of basin is about 332.5 sq.kms with very coarse drainage texture and the region has high resistant of highly permeable sub soil materials, dense vegetation cover and low relief.

Keywords: Remote Sensing, Geographic information system, Watershed and Morphometric parameters.

Introduction:

Water resources on land are very unevenly distributed both spatially and temporally. Idiosyncrasies of monsoon and diverse physiographic conditions give rise to unequal distribution of water. Over the years, increasing population, urbanization and expansion in agriculture has accentuated the situation. The dependency on the groundwater is expected to increase

in future and the requirement in 2050 is estimated about more than three times to present level due to increase in population (Gupta and Deshpande, 2004). The aftermath of unscientific exploitation of groundwater is that we are moving towards water stress condition. Despite being a very important part of the nation's growth, water resources analysis has been fragmentary. An integrated study covering the aspect of

watershed management and groundwater recharge is a crucial requirement of the present day (Choudhury, 1999).

Remote sensing and GIS are playing a rapidly increasing role in the field of hydrology and water resources development. Information from satellites is becoming more and more important for environmental research. An important part of this information concerns water – an element most essential for man, its phases and peculiarities. Remote sensing provides multi-spectral, multi-temporal and multi-sensor data of the earth's surface (Choudhury, 1999). One of the greatest advantages of using remote sensing data for hydrological investigations and monitoring is its ability to generate information in spatial and temporal domain, which is very crucial for successful analysis, prediction and validation (Saraf, 1999). However, the use of remote sensing technology involves large amount of spatial data management and requires an efficient system to handle such data. The GIS technology provides suitable alternatives for efficient management of large and complex databases. Watershed monitoring and management has been found to be economical and faster with the usage of the capabilities of a GIS. Erosion and sediment yield from the watershed can be assessed using a suitable model in raster GIS (Garg, 1991). On the basis of erosion and other environmental parameters, watershed prioritization can be carried out in GIS. In recent time many workers such as Raghuvansi et al., 2008; Prasad et al., 2008; Rokade et al., 2007; Jha et al., 2007; Leblanc et al., 2007; Saraf et al., 2004 and Teeuw, 1995 used GIS for evaluation and management of groundwater resources of the various terrains. One of the greatest advantages of using remote sensing data for groundwater investigation and monitoring is its ability to generate information in spatial and temporal domain, which is very crucial for successful analysis, prediction and validation (Jaiswal et al., 2003; Murthy, 2000; Saraf and Choudhury, 1998; Krishnamurthy et al., 1996).

A watershed, which is an area from which runoff resulting from precipitation flows past

a single point into large stream, river or lake is taken as a unit for the study of drainage characteristics. Drainage system refers to the origin and development of streams through time. The location number and flow directions of different streams of a drainage system depends on the nature of slope, structural control, lithological characteristics, tectonic factors, climatic characteristics, vegetal characteristics etc. Geology, relief and climate are the primary determinants of running water ecosystems functioning at the basin scale (Mesa, 2006). Detailed morphometric analysis of a basin is a great help in understanding the influence of drainage Morphometry on landforms and their characteristics (Sreedevi P.D et al., 2009). Thus the study of drainage characteristics through morphometric analysis (quantitative description of the drainage basin geometry) of a watershed in a region gives much information regarding the denudational history, subsurface material, geological structure, soil type and vegetation status of that region, which play a crucial role in formulating a plan for watershed management.

The present work is an attempt towards this direction. The study focuses on development of remote sensing and GIS based analysis and methodology in watershed modelling for conservation and management of water resources. In order to demonstrate the integrated remote sensing and GIS based methodology, an area which forms Krishnamgaripalli watershed situated in the state of Andhra Pradesh (India) has been taken.

Study Area:

The area lies between Pulivendula and Sanivaripalli situated between parallels of $78^{\circ}00'0''$ to $78^{\circ}15'0''$ E longitude and $14^{\circ}15'0''$ to $14^{\circ}30'0''$ N latitude with intended boundary falling in Survey of India (SOI) topographic sheet # 57J 03 on 1:50,000 scale, covering an area of over 728 sq. kms. The study area includes five revenue mandals namely, Pulivendula, Lingala, Udumakurti, Krishnamagaripalli, Sanivaripalle are the important towns in the area. The central and southern part of study

area is occupied by high hills, ridges and valleys. The Maddaleru is the only river in the study area which is a seasonal river fed by monsoon (June -October), which is flowing in the NW direction in the southern part of the imagery. Some major tanks also

cover in the same area. The climate is dry with mean annual rainfall of 100-150 cm and mean annual temperature of 32°C. The value of contours in the study area ranges from 200-600m. The slope category is within 1-15 degrees.

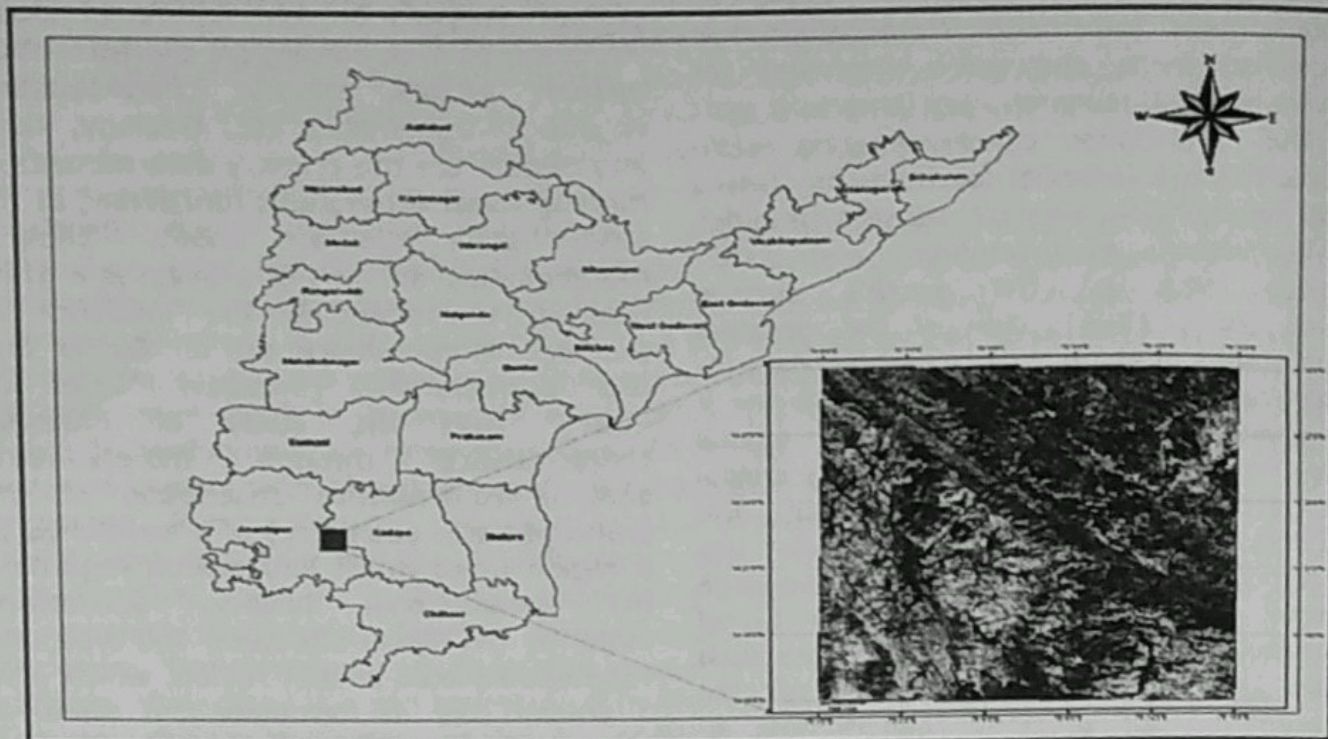


Figure 1: Map Showing Location of the Study Area

Objectives:

The objectives of Watershed Development Projects are:

- To study the present status of water resources, land resources, cropping patterns, forest cover etc., using satellites data, collateral data and field data.
- To suggest suitable sites and methods for watershed management to augment groundwater recharge in the area.
- To develop and test an integrated remote sensing and GIS technique in watershed planning and groundwater recharge investigations.

Data Used and Methodology:

The study used a satellite geocoded imagery (IRS P6 LISS-III), acquired June 2001 with spatial resolution of 30 meters, Survey of India toposheet 57 J 03 on 1:50,000 scale and tracing sheets. The digital data were processed for geometric and radiometric corrections using ERDAS image processing

software. A number of softwares such as Arc GIS version 9.0, ERDAS Imagine version 9.1 and Microsoft office processing tools (Word, Excel) are used in the study. For processing ERDAS was considered useful. The study was applied both visual and digital Landsat image interpretation in order to generate various thematic maps respectively. The field equipments and tools used for fieldwork include Brunton compass, Garmin e Trex H (GPS) receiver, slope meter, digital camera, soil knife, hammer, and measuring tape. The main methodology of the research is to prepare thematic maps with the application of remote sensing and GIS software. The actual work involved a comprehensive gathering of information and linked to the related activities. The development and planning of such a vast area for a proposed project requires various data information from the field to identify the terrain characteristics, etc. This is composed of getting use of space images and GIS, complemented with truthing fieldwork as

well as making use of the available information from several concerned agencies and sectors in the study area. The flow chart showing the methodology of land resources analysis is given in fig.2.

Preparation of Base Map:

The base map is the essential requirement for any mapping using remote sensing techniques or by ground methods. It is the map which is used for collection and representing the thematic (resource) information on a common uniform scale. In the study area we used SOI toposheet No. 57 J 03 for the purpose of base map preparation. The major features like important town or villages, rivers, reserved forest area and transport network were traced on tracing sheet on the basis of which the base map has been prepared on 1:50,000 scale.

Preparation of Thematic Maps:

These maps are the true representation of earth's phenomena such as spatial distribution of natural resources existing at the time of survey (Ravi Gupta, 2003). In the present study satellite image (IRS P6 LISS-III) which is a true record of the various environmental resources information on the base map. These maps showing spatial distribution of forest, agriculture, soil, water resources etc., are prepared by visual interpretation of the satellite imagery. Visual interpretation is carried out based on the image characteristics like tone, size, shape, pattern, texture etc. in conjunction with existing map/literature. These pre-field thematic maps are modified substantiated and confirm after limited field checks. The image interpretation keys used for land use land cover classification are shown in table 1 (Lillisand and Kiefer, 1979; Pandey, 1990).

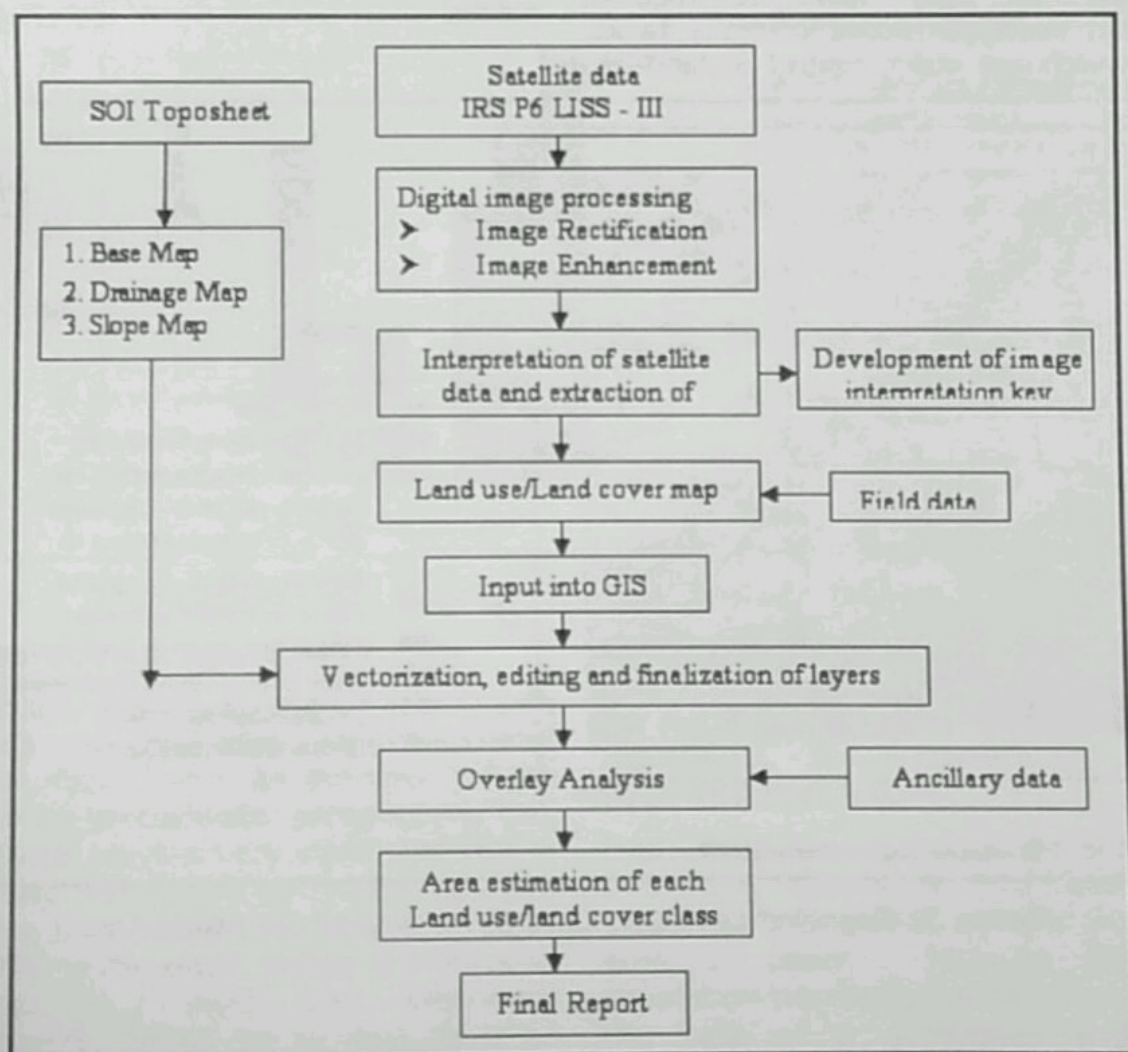


Figure 2: Flow Chart Showing the Broad Methodology Adopted In the Study

Results and Discussions:

A watershed is a geo-hydrological unit, which drains into common point. The watershed approach is a project based, ridge to valley approach for in situ soil and water conservation, afforestation etc. Unit of development will be a watershed area of about 500 ha. The thematic maps generated from satellite data for different themes such as land use/land cover, hydro geo - morphology, soils etc. may be used for selection of a watershed area. The project will primarily aim at treatment of non-forest wastelands and identified drought prone and desert areas.

Water management has increasingly been realized to be an essential component of sustainable water development. While no-one argues with the desirability of proper watershed management, achieving it has not been an easy task. It requires concurrent accomplishment of many tasks, among which are strict control of land-use

practices, afforestation and forest management, and implementation of appropriate soil and water conservation practices.

Thematic Maps Prepared for the Demarcation of the Watershed:

1. Preparation of Land Use - Land Cover Map:

The knowledge of land use and land cover is very important for land planning and land management activities (Anji Reddy, 2002). The land use / land cover system adopted in mapping of earth surface features is a system derived from the United States Geological Survey (USGS) land use / land cover classification system. The Land use / Land cover map is used for the easy demarcation of watershed in the much needed location. Land use / land cover map prepared using satellite imagery is shown below.

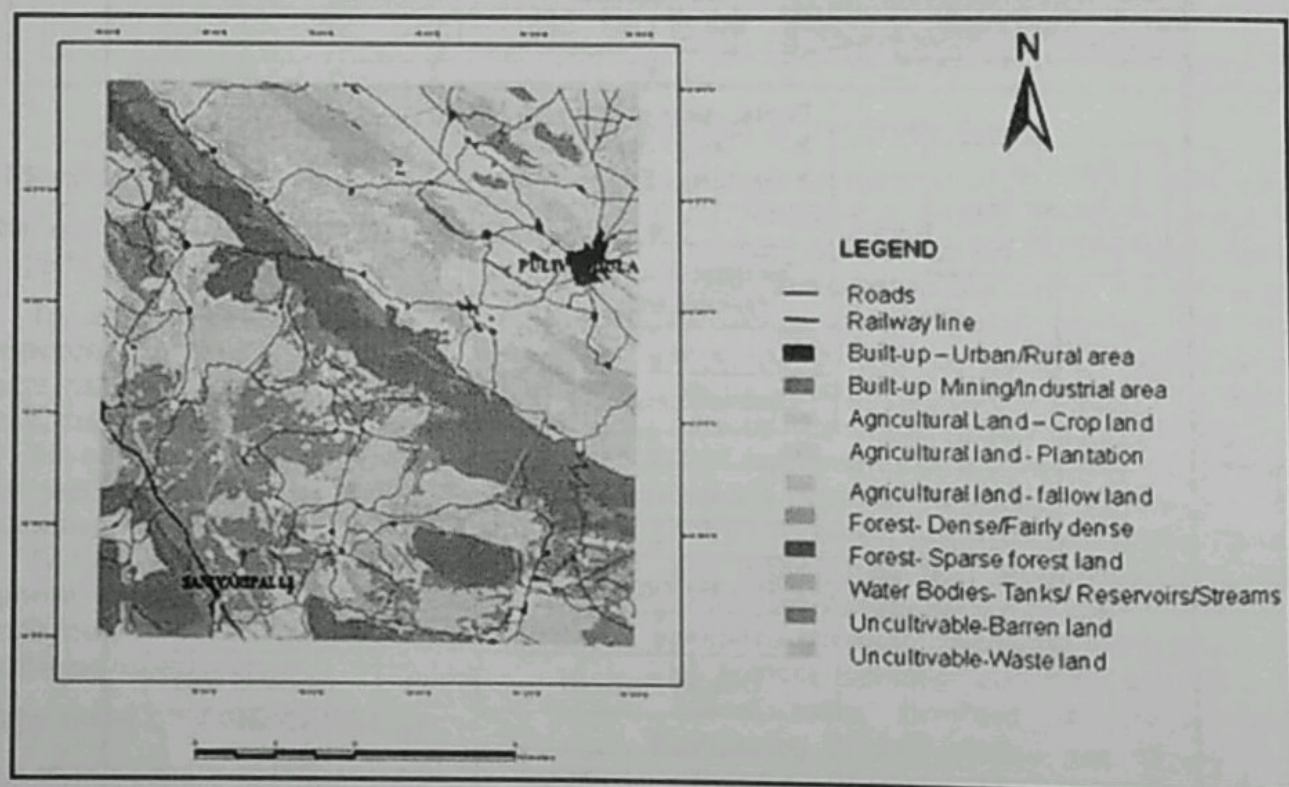


Figure 3: Classified Land Use/Land Cover Map of the Study Area

Table 1: Table Showing Land Use / Land Cover Statistics of the Study Area

S.No	DESCRIPTION	Area in Ha	Percent to total study area
1	Built Up area (Urban)	316.13	0.43
2	Built Up area (Rural)	484.08	0.65
3	Built Up-Mining / Industrial area	219.02	0.29
4	Agricultural Land-Crop Land	14820.71	19.43
5	Agricultural Land-Plantation	2679.1	3.61
6	Agricultural Land-Fallow-Current Fallow	6089.8	8.85
7	Forest-(Dry/Moist/Thorn)-Dense/Closed	264.15	0.36
8	Forest-Scrub Forest	12232.58	16.47
9	Uncultivable-Barren Rocky/Stony waste land	15708.55	21.16
10	Uncultivable-Wastelands-Scrub land	18792.1	25.31
11	Waterbodies-River/Stream/ Reservoir/Tanks	2554.3	3.44
	Grand Total	74250.46	100

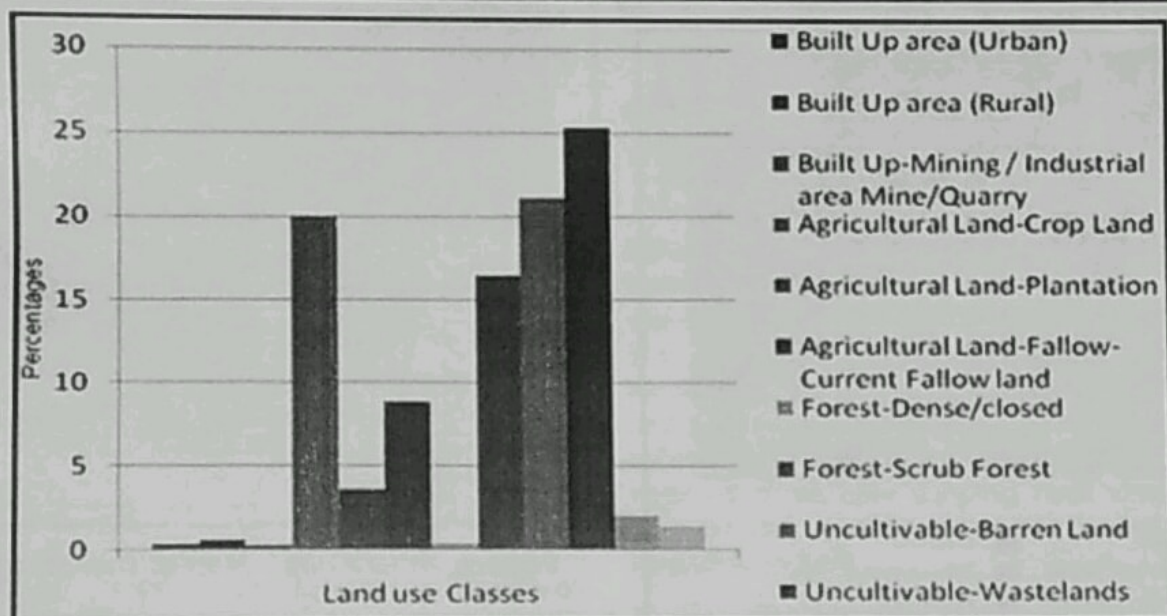


Table 2: Bar Chart Showing the Percentage of Land Use Classes

2. Preparation of Slope Map:

Slope of any terrain is one of the factors controlling the infiltration of groundwater into subsurface, hence an indicator for the suitability for groundwater prospect (Horton, 1932). Slope plays a very significant role in determining infiltration vs. runoff relation. Infiltration is inversely proportional to slope i.e., gentler is the slope, higher is infiltration and less is runoff and vice-versa (Montgomery, 2006). Slope analysis is an important parameter in geomorphic studies. The slope elements, in turn are controlled by

the climatomorphogenic processes in the area having the rocks of varying resistance. The input for the preparation of slope map is toposheet. The procedure for making a slope map is as follows.

First, mark the horizontal distances for all the slope classes as per the scale of toposheet and specific contour interval on a strip of paper. Measure the contour separation perpendicular to the contour with the help of a graduated strip already prepared and assigns slope class to the space between the contours. Verify the

break in slopes with the help of toposheets. Demarcate the slope classes with pencils of appropriate colors on tracing paper overlay on the toposheet. Table 1 shows the slope angle, horizontal distance and colour scheme delineated on toposheet of 1:50,000 scale with 20m contour interval.

Table 3: Showing the Slope Angle Classification and Colour Index

Slope angle	Horizontal Distance (cm)	Color
0-2	>11.5	Yellow
5-Feb	11.5-4.5	Green
15-May	4.5-1.5	Orange
15-25	1.5-0.85	Pink
25-35	0.85-0.57	Red
35-55	0.57-0.28	Brown
>55	<0.28	Purple

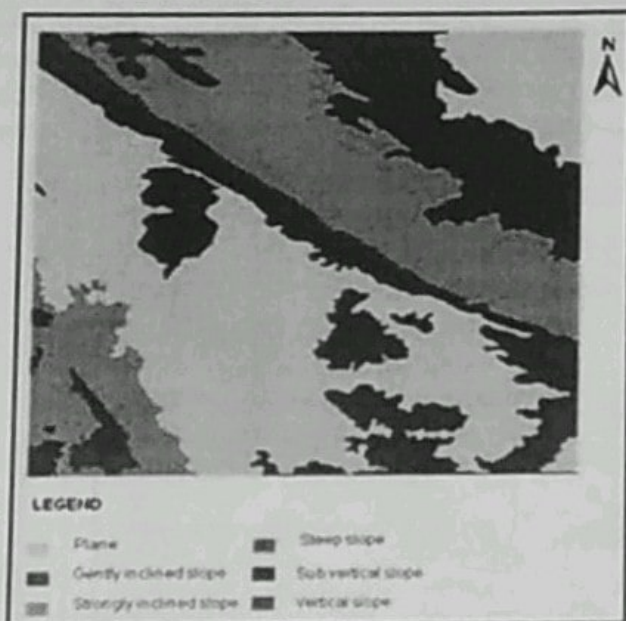


Figure 4: Showing the Slope Map of the Area

In the study area, the south eastern part have strong inclined slopes in areas of Sanivaripalli situated in north west corner, Lakkasamudram (central part) and Udukulakurti and Siddugaripalli in north east corner. Whereas total north eastern part is having slope angle decreasing from 5-1 degrees i.e., strongly inclined to plane towards NE direction. As per the criteria of watershed demarcation the area should have a slope such that the gradient should decrease from higher angle to lower angle.

3. Preparation of Drainage Map:

A drainage map based on the SOI toposheet (1:50,000 scale) and satellite imagery was prepared using ARC GIS-module. The number of streams present in the area is demarcated as shown in the adjacent map.

The water category includes streams, lakes, canals and reservoirs. The major tanks/reservoirs in the study area present around Lakkasamudram, Reddivaripalli, Sanivaripalli, Gollapalle etc in Southern part of imagery only. Streams, canals are generally located in Northern part of imagery.

Maddaleru river is flowing towards NW direction in the Southern Western part of imagery. The villages adjacent to the river are Sanivaripalli, Adavibrahmanpalli, Nakkalapalli, Dorigallu, Indukuru and Maartadu.



Figure 4: Drainage Map of the Study Area

4. Watershed Demarcation and Morphometric Analysis:

After studying the total conditions of the area it is concluded that, the region is very poor in water resources, top priority should be given for its development. The ground water distribution in the study area is not uniform and it depends on the drainage pattern. For this watershed, near the village Krishnamgaripalli is demarcated so as to fulfill the needs. The watershed area is represented between Eguvapalli in the west and Krishnamgaripalli in the east. The demarcated watershed is named after Krishnamgaripalli village. The watershed

covers an area of 332.5 sq. km (Fig. 4). It enjoys a tropical climate with rainfall between 100-150cm annually.



Morphometric Analysis:

Morphometry is the measurement and mathematical analysis of configuration of Earth surface, shape and dimensions of its landforms. The main purpose of the work is to discover holistic stream properties from the measurement of various stream attributes. The quantitative drainage analysis is done aspect wise such as linear aspects and aerial aspects.

Linear Aspects:

i. Stream Order (U):

The first step in drainage basin analysis is designation of stream orders. The stream order is a measure of the degree of stream branch within a watershed. Each length of stream is indicated by its order for example first order, second order and third order etc. In the given watershed 4 stream orders have been calculated.

ii. Stream Number (Nu):

The count of stream channel in its order is known as stream number. The number of streams decreases as the stream order increases. The demarcated watershed has the following stream orders and stream number.

Stream order	Stream number
First order	100
Second order	20
Third order	6
Fourth order	1

iii. Bifurcation Ratio (Rb):

Bifurcation ratio is defined as the ratio of the number of streams one order to the number of the next higher order. It is given as below:

$$\text{Bifurcation ratio (Rb)} = \text{Nu} / (\text{Nu}+1)$$

No of order	Bifurcation ratio (Rb)
First order	4.76
Second order	2.85
Third order	3

Bifurcation ratios characteristically range between 3 and 5. This indicates that in watershed geologic structures do not distort the drainage pattern. The hypothetical minimum value of 2 is rarely approached under natural conditions. Abnormally higher bifurcation ratios might be expected in regions of steeply dipping rock strata where narrow strike valleys are confirmed between hogback ridges.

iv. Stream Length (Lu):

Mean length of a stream channel segments of order U is a dimensional property revealing the characteristic size of components of a drainage network and contributing basin area.

No of order	Stream length (Lu)
First order stream	98.75
Second order stream	33.25
Third order stream	29.75
Fourth order stream	10.25

v. Stream Length Ratio (RL):

Stream length ratio is the ratio of main length (L_u) of segments of order U to main length of segments of the next lower order (L_{u-1}). Here the stream length ratio tends to be constant throughout the successive orders of a watershed. It indicates that stream lengths are decreasing with increase in the order of stream.

$$RL = L_u / (L_{u-1})$$

No of order	Stream length ratio (RL)
Second order stream	1.85
Third order stream	1.66
Fourth order stream	2.06

Aerial Aspects:**vi. Drainage Area:**

Drainage areas are defined as the area which contributes water to a particular channel or set of channels. The basin area is one of the important parameters like that of the length of the draining the basin. In the demarcated watershed the total drainage area or basin area is about 332.5 sq.km.

vii. Drainage Net:

Drainage net is related to the hydraulic geometry of stream channel. If the streams are relatively far apart then, the drainage net is less dissected. If the streams are relatively low spaced then the drainage net is intensively dissected. Here in the Krishnamgaripalli watershed the drainage net is less dissected because the streams are relatively far apart.

viii. Drainage Density (Dd):

Drainage density is the total length of the all streams in the basin to the area of whole basin. It is calculated as here under:

$$Dd = \epsilon L_u / A_u$$

$$= 162.95 / 332.5$$

$$= 0.490 \text{ km/sq. km}$$

The low values of drainage density indicates the region is high resistant of highly permeable sub soil materials dense vegetation cover and low relief.

xi. Drainage Texture:

Drainage texture is the measure of the closeness of the channel spacing and related directly to the drainage density. The drainage density values of the watershed are 0.490 km/ sq. km. i.e., less than two. Therefore the drainage areas have very coarse drainage texture.

Table 4: Morphometry of Krishnamgaripalli Drainage Basin

Morphometric parameters	Stream Order (U)			
	I	II	III	IV
Stream number (N_u)	100	20	6	1
Stream length (L_u) km	98.75	33.25	29.75	10.25
Bifurcation ratio (R_b) = N_u / N_{u+1}	4.76	2.85	3	-----
Stream length ratio $RL = L_u / L_{u-1}$	-----	1.85	1.66	2.06
Total drainage area (A_u)	332.5 sq.km.			
Average Bifurcation ratio (R_b)	3.6695			
Drainage density, $Dd = \epsilon L_u / A_u$	$162.95 / 332.5 = 0.490 \text{ km/sq. km}$			
Drainage Texture	0.49 km/ sq. km. (<2), Very coarse drainage texture.			

Conclusion:

From the above discussions it is concluded that Remote sensing and Geographical Information system (GIS) are very effective tools in the study of drainage characteristics in a region. Different components of

watershed, such as drainage, physiography, land use etc., through remote sensing data in combination with SOI topographic sheets help to assess their potential land use. The total drainage area of basin is about 332.5 sq.kms with very coarse drainage texture and the drainage net is less dissected

because the streams are relatively far apart. Bifurcation ratios characteristically range between 3 and 5 for watershed in which geologic structures do not distort the drainage pattern. Low values of drainage density indicates that the watershed region is high resistant of highly permeable sub soil materials dense vegetation cover and low relief. These Guidelines should be taken as general principles for implementation of watershed development projects and should not be used as a tool to make their implementation a complex exercise. The basic philosophy of these general principles is to seek gainful and transparent utilization of public funds for watershed development, with a view to promoting the overall economic development and improving the socio-economic condition of the resource poor and the disadvantaged sections of the people inhabiting the project areas.

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